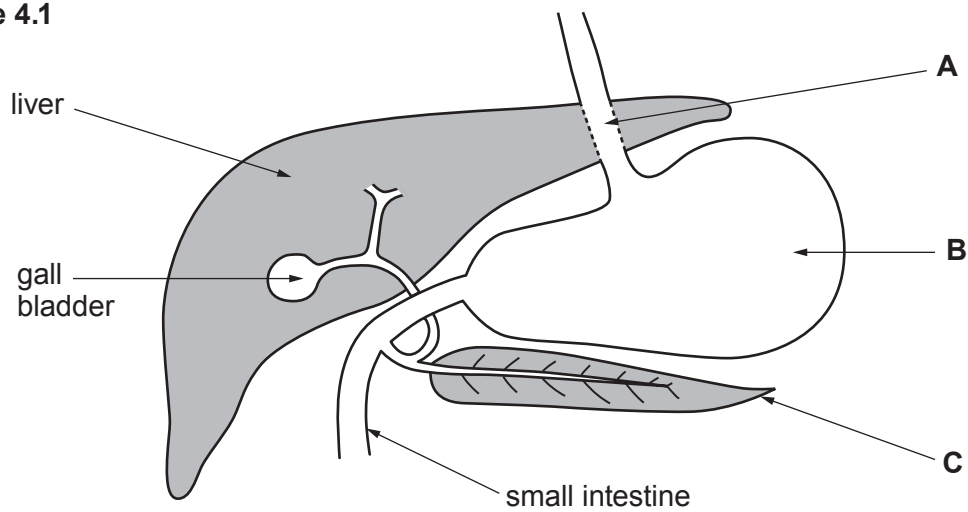


4. **Image 4.1** shows part of the digestive system in a human.

Image 4.1



(a) (i) State the letter (**A-C**) which shows the pancreas. [1]

.....

(ii) The pancreas produces lipase.

Complete the sentence below using one of the terms given.

glucose glycerol amino acids

Lipase breaks down fat into fatty acids and [1]

(iii) The breakdown of fat by lipase is assisted by bile.
Using **Image 4.1** describe the pathway taken by bile as it passes from the **liver**,
where it is produced, to where it enters the **small intestine**. [2]

.....
.....
.....
.....



- (b) In an investigation, the effect of lipase on the breakdown of fat was measured with and without using bile. Five trials were carried out and the results are shown in **Table 4.2** and **Graph 4.3**.

Table 4.2

		Lipase activity (a.u.)	
		With bile	Without bile
Trial	1	24.9	3.1
	2	22.6	2.3
	3	26.2	4.4
	4	22.6	1.7
	5	23.7	3.5
Mean		24.0	3.0
Ratio of means = :			

- (i) I. Calculate the ratio of the means. **Write your answer in Table 4.2.** [1]
II. Complete **Graph 4.3** by **plotting the results** for trial **4** and trial **5** from **Table 4.2.** [3]

Graph 4.3

